

Supplementary Material

How Predictable Are Rare Precipitation Extremes? A Calibration-Aware Assessment for London and Paris

Fankai Dong

Olatunji Johnson

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S1 Results at the 99th-percentile threshold

The main paper analyses the 95th-percentile index, which has 417 extreme days (287 train, 130 test). The 99th-percentile index is far scarcer — only 137 extreme days in thirty years (101 train, 36 test) — which precludes the held-out intensity analysis (that requires extreme days in both train and test across several splits). For completeness we report here the *occurrence* results of the simpler models at the 99th percentile; the deep Gaussian-process extreme-value hurdle, already dominated at the 95th percentile, is not re-run.

Table S1 shows the bake-off (seed-averaged over three seeds, horizon $h = 1$), scored against climatology as in the main paper. The qualitative picture matches the 95th percentile but is weaker and noisier, as expected with a third as many events:

- **Occurrence remains weakly predictable.** A plain calibrated logistic regression with the ERA5 drivers beats climatology (Brier skill score +0.015), improving on index history alone (+0.001); the gain again comes chiefly from the convective-instability and circulation fields. Resolution (informativeness) is positive but smaller than at the 95th percentile (0.36×10^{-3} versus 0.83×10^{-3}).
- **The distributional models fall below climatology.** With only 101 training exceedances, the zero-inflated gamma and eGPD regressions are slightly worse than climatology on the Brier score, and the zero-inflated log-normal is again catastrophically miscalibrated — echoing, more sharply, the main-paper finding that the response distribution matters and that data scarcity penalises the more flexible models.
- **The tail is climatology-bound.** No model beats the climatological tail on the threshold-weighted CRPS (all skill scores ≤ 0), and the scarce exceedances do not support the held-out intensity test.

In short, the 99th-percentile threshold tells the same story as the 95th — weak but genuine occurrence predictability led by convective instability, a climatology-bound tail, and a penalty on over-flexible models — but with too few events to support the intensity analysis or to add to the main conclusions.

S2 Paris: per-city diagnostic figures

The main paper shows the data, calibration, and horizon-decay diagnostics (Figures 1, 2, and 4 there) for London as the representative city, while the core comparative results — the bake-off, the held-out intensity analysis, and the unified hurdle — are reported for both cities. For completeness we give the Paris counterparts of the three single-city diagnostic figures here. They are qualitatively identical to London: a zero-inflated, right-skewed, short-memory index with a

Table S1: Bake-off at the 99th-percentile threshold (London, $h = 1$, seed-averaged). Brier skill score (BSS) and threshold-weighted CRPS skill (twCRPSS) are relative to climatology; positive is better. RES is the Brier resolution ($\times 10^3$). Climatology and a stationary peaks-over-threshold model define the reference. Compare with Table 2 of the main paper (95th percentile).

Model	BSS	RES $\times 10^3$	twCRPSS
Climatology / stationary POT	0.000	0.00	0.000
Logistic (index history)	+0.001	0.08	—
Logistic (+ERA5)	+0.015	0.36	—
ZI gamma (+ERA5)	−0.069	0.20	−0.130
ZI extended-GPD (+ERA5)	−0.048	0.26	−0.087
ZI log-normal (+ERA5)	−1.153	0.25	−4.827

summer convective peak (Figure S1); well-calibrated exceedance probabilities from the simple logistic model, with the deep Gaussian-process hurdle no better calibrated (Figure S2); and occurrence skill decaying toward the prevalence floor with lead time while conformal coverage stays at the nominal level (Figure S3). Paris differs from London only in having a higher extreme-day prevalence (a more convective regime), consistent with its stronger predictability in the main text.

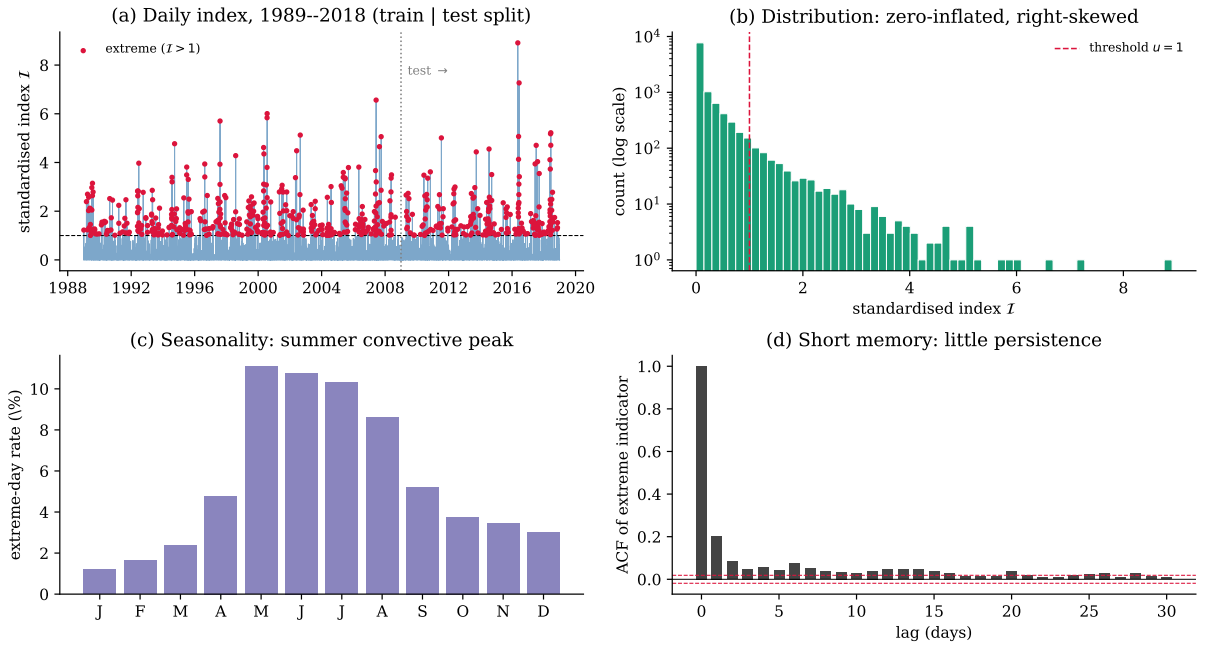


Figure S1: Paris 95th-percentile precipitation-extreme index (Paris counterpart of Figure 1 in the main paper). **(a)** Daily index with extreme days ($I > 1$) and the train|test split; **(b)** zero-inflated, right-skewed distribution; **(c)** summer-peaked seasonality of the extreme-day rate; **(d)** short memory of the extreme indicator.

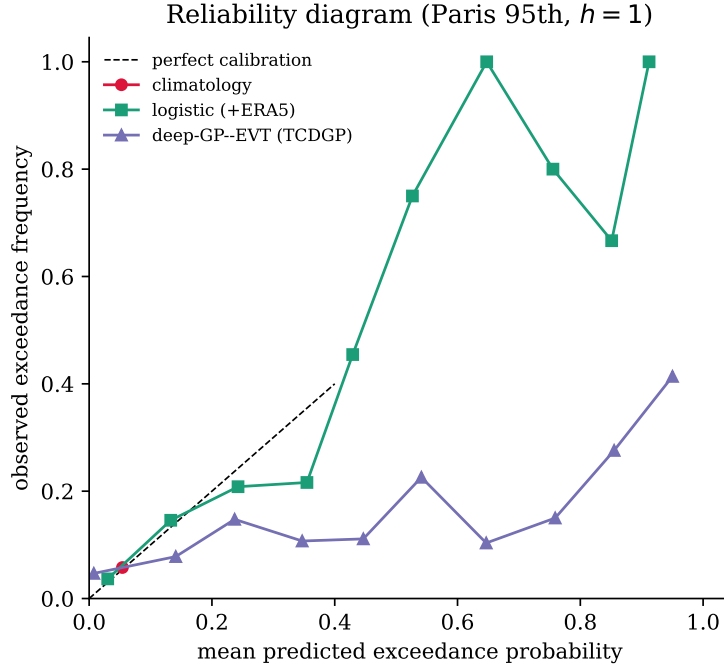


Figure S2: Reliability diagram for the Paris exceedance probability (95th, $h = 1$; Paris counterpart of Figure 2). The calibrated logistic model tracks the diagonal; the deep Gaussian-process extreme-value hurdle is no better calibrated, mirroring London.

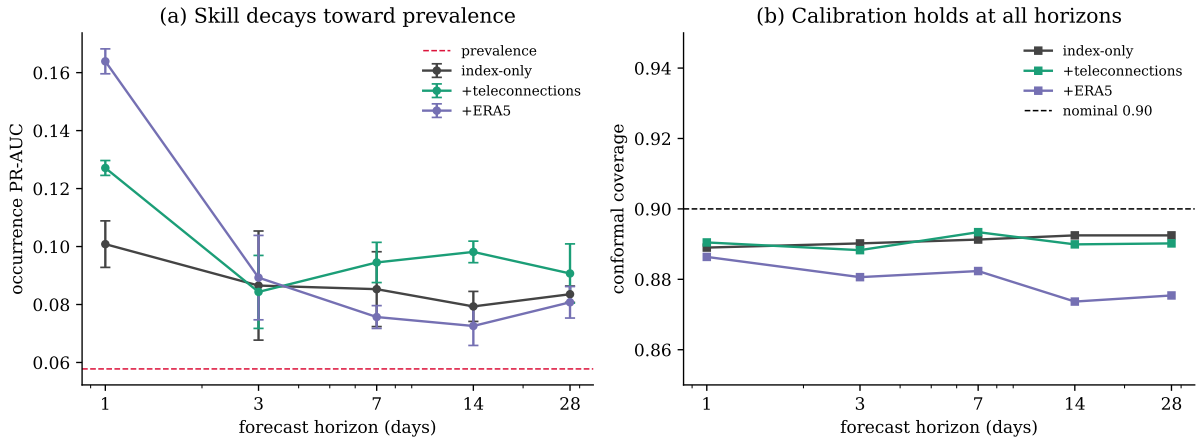


Figure S3: Seed-averaged horizon sweep on the Paris 95th-percentile index (Paris counterpart of Figure 4). **(a)** Occurrence PR-AUC decays toward the prevalence floor (dashed) as the horizon grows, with marginal driver gains; **(b)** conformal coverage holds at the nominal 0.90 across horizons and predictor sets. Markers are means over three seeds.